

Albert Wang

aw3741@columbia.edu | [albert-wang1010.github.io](https://github.com/albert-wang1010) | [linkedin.com/in/albert-wang1](https://www.linkedin.com/in/albert-wang1)

EDUCATION

Stanford University

M.S. Electrical Engineering, Integrated Circuits

Columbia University & Bates College | 3-2 Dual Degree Program

B.S. Electrical Engineering (Columbia) | GPA: 3.72/4.0 | Dean's List 2024{2026

B.A. Physics, Minor in Philosophy (Bates) | GPA: 3.97/4.0 | Dean's List 2021{2024

Sept. 2026 – Feb. 2028

Stanford, CA

Aug. 2021 – May 2026

New York, NY

Lewiston, ME

TECHNICAL SKILLS

IC Design: CMOS transistor sizing, op-amp/filter design, small-signal analysis, biasing, stability; Cadence Virtuoso (schematic/layout/DRC/LVS/extraction), LTspice, Synopsys Design Compiler, PrimeTime

HDL & Digital: Verilog, VHDL, SystemVerilog; RTL/gate-level simulation, timing analysis, FSM design, synthesis

Lab & PCB: Oscilloscope, function generator, DMM, soldering; Altium, Autodesk Fusion 360, Autodesk Revit

Programming: Python, C/C++, MATLAB, Java; signal processing, data analysis

Languages: English (native), Chinese (fluent)

PROJECTS

Two-Stage CMOS Op-Amp with Miller Compensation

Fall 2025, New York, NY

- Designed two-stage OTA with Miller compensation: gain = 4 V/V, 76 dB open-loop, 47° phase margin, 203 kHz GBW.
- Optimized transistor sizing (W/L 4{184) for 700 μ A budget; <0.1% DC accuracy, 5.4 μ s settling time, <6.5% overshoot.
- Performed AC/DC/transient analysis in Cadence Spectre; verified stability across PVT corners.

Full-Custom 65nm VLSI Microprocessor

Fall 2025, New York, NY

- Designed complete 8-bit microprocessor schematic-to-layout in 65nm CMOS using Cadence Virtuoso; datapath includes ALU, shifter, 8 $\times$ 8 SRAM, decoder, and PLA control unit.
- Achieved 100 MHz with 88.8% timing margin, 4,100 μ m² area; completed full DRC/LVS verification; documented parasitic characterization (2{3 $\times$ delay increase) in 18-page technical report.
- Post-layout power optimization: 876 μ W total consumption; analyzed process, voltage, and temperature (PVT) sensitivity.

CLACS Smart Pen – Senior Design Project

Spring 2026, New York, NY

- Designed LiPo battery/USB-C charging subsystem and revised STM32WB55 MCU schematic in Altium; co-developed double-sided PCB with ~84% area reduction.
- Hand-soldered SMD boards (IPC-J-STD-001); wrote I2C firmware that isolated solder-bridge and USB-C routing faults.
- Wrote full C middleware for live IMU streaming over USB-CDC/BLE 5.4; adaptive stroke detection improved segmentation ~3 $\times$ over baseline.

64-Tap FIR Filter with Clock Domain Crossing

Fall 2025, New York, NY

- Designed 64-tap 16-bit FIR filter in Verilog with asynchronous FIFO, MAC with saturation, FSM controller.
- Synthesized using Synopsys Design Compiler (IBM 130nm) achieving 100 MHz timing closure, 46,551 μ m² area; validated via gate-level simulation with SDF back-annotation.
- Power analysis via PrimeTime PX: 3.384 mW total, 0.034 mW/MHz efficiency, 100% VCD annotation.

EXPERIENCE

Hardware Security Intern (PUF/HSM)

July 2025 – September 2025

eMemory Technology Inc.

Hsinchu, Taiwan

- Co-authored internal API specification for PUF-backed HSM framework covering PKCS#11 partitions, HA/backup, and operations; document adopted as reference for FIPS 140-3 certification testing.

Collegiate MEP Intern

May 2025 – July 2025

PBK Architects (LEAF Engineers)

Houston, TX

- Developed NEC-compliant panel schedules and circuiting for large-scale commercial and institutional building electrical systems using Autodesk Revit.
- Performed load calculations, feeder/breaker sizing, and phase balancing; standardized circuit naming conventions to improve drawing consistency across design reviews.

RESEARCH

NSF-REU Computational Physics Researcher

May 2024 – October 2024

Clarkson University

Potsdam, NY

- Developed projection-based learning algorithm in MATLAB achieving 100 $\times$ speedup for photonic crystal simulations.
- Co-authored peer-reviewed paper at SPIE Photonics West 2025 (DOI: 10.1117/12.3028208).

Undergraduate Researcher, Neutrino Detection

May 2023 – May 2024

University of Massachusetts Amherst

Amherst, MA

- Investigated SiPM photon detection efficiency in liquid xenon for nEXO neutrinoless double beta decay experiment; developed waveform analysis pipeline in Python/Geant4; presented at APS April Meeting 2024.